

# Activation Procedures

Activation procedures are provocative maneuvers that can elicit epileptiform abnormalities in susceptible individuals. Two commonly employed activation procedures are hyperventilation and photic stimulation. Prior sleep deprivation may be used to increase the yield of an EEG but isn't considered a "maneuver."

## Hyperventilation (HV)

### Procedure

The patient is asked to take deep regular breaths at 20–30 times per minute for about 3–5 min (or sooner if they wish) and the EEG recording is continued for at least 3 min afterward. Balloons or toy windmills are helpful in children.

### Contraindications

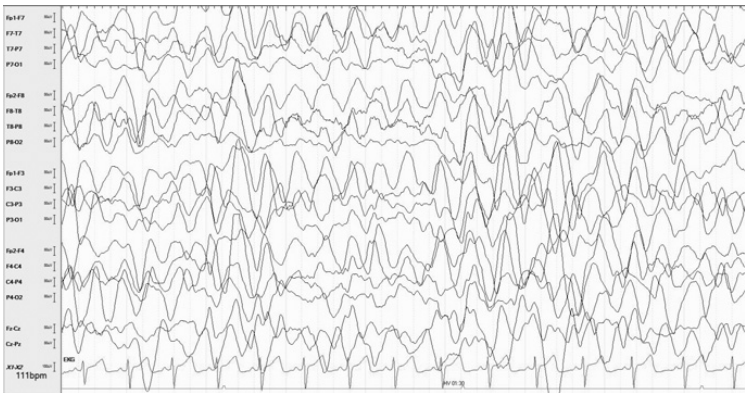
Cerebrovascular, cardiac, and pulmonary diseases, sickle cell disorder, and Moya-Moya disease are important contraindications.

### Description

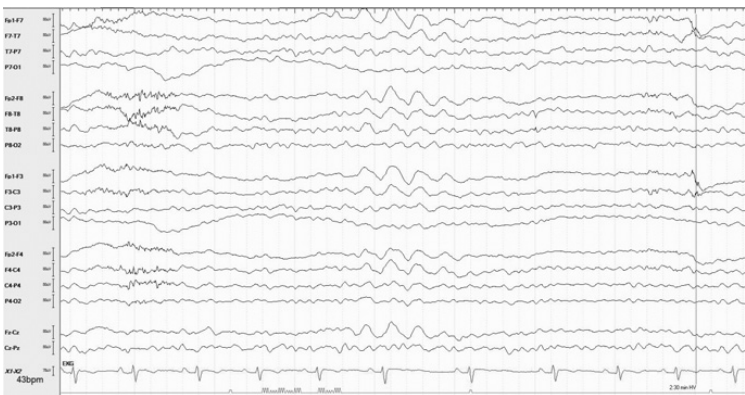
The effort, clinical effects (symptoms, seizures, or nonepileptic attacks) and the EEG response are noted. Sinus tachycardia is commonly seen in the single EKG channel; Hyperventilation must be stopped if ST segment changes or arrhythmias are noted.

### Normal Electrographic Response

A symmetric high-amplitude generalized background slowing with a frontal predominance is typical in children and young adults (called buildup). This response is age dependent and may be absent in older individuals. Figure 17.1 shows a prominent HV related build up in a child, Figure 17.2 shows a milder response in an adolescent.



**Figure 17.1** Five-year-old boy undergoing HV, symmetrical high-amplitude generalized delta slowing characteristic of a normal buildup.

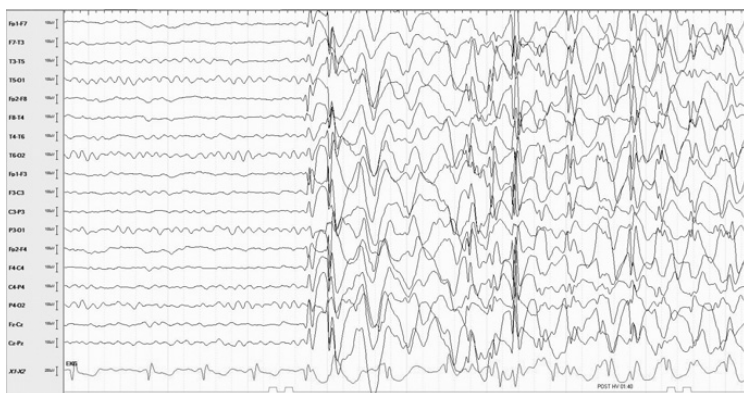


**Figure 17.2** Seventeen-year-old girl undergoing HV, comparatively less prominent buildup.

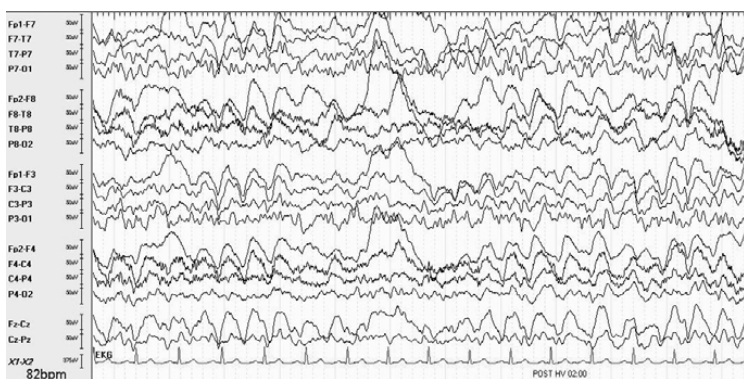
## Abnormal Responses

1. Seizures (commonly absences) or epileptic discharges may be elicited in certain generalized epilepsies such as childhood absence epilepsy (CAE). Figure 17.3 shows an absence seizure triggered after HV.

2. HV-induced build up should typically disappear as soon as normal breathing is resumed. Prolonged build up that persists beyond 1 min of stopping hyperventilation is nonspecific but may be associated hypoglycemia (resolves after a snack) or cerebrovascular disease such as Moya-Moya. Figure 17.4 shows prolonged hyperventilation build up associated with Moya-Moya disease.



**Figure 17.3** Twelve-year-old boy with CAE, absence seizure-induced post HV.



**Figure 17.4** Nineteen-year-old woman with Moya-Moya disease, prolonged HV buildup.

3. Asymmetric response or focal slowing may be seen in older individuals with atherosclerotic cerebrovascular disease [1,2].

## Photic Stimulation (PS)

Intermittent photic stimulation is best performed a few minutes after hyperventilation. This technique is useful in diagnosing photosensitive epilepsies and tailoring individual treatments.

## Procedure

A series of 5 to 10 s blocks of photic stimulation are administered using a flashing lamp in a dimly lit room. At our institution, we start with a flash frequency of 1 Hz and increase in increments of 2 Hz to maximum of 30 Hz. The procedure should be stopped as soon as epileptiform discharges are seen to prevent provoking a seizure.

## Contraindications

Photic stimulation is routinely performed as part of a standardized electroencephalogram. The risk of a provoked generalized tonic clonic seizure is very low (0.04%) [3].

## Description

Any clinical effects such as myoclonus, absences, eyelid fluttering, and headaches should be recorded along with the flash frequency and EEG response. Nonepileptic attacks are also common.

## Normal Electrographic Response

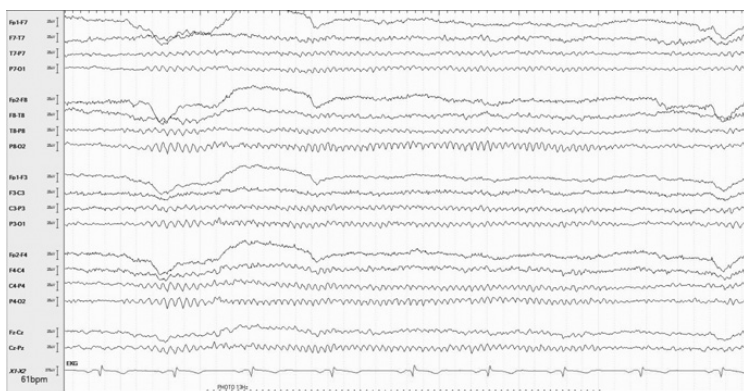
Symmetric photic driving response is characterized by repetitive sharp, positive waves that occur with an occipital dominance at the flash frequency or a slower harmonic. A fragmented or absent response is also normal. Figure 17.5 shows a normal photic driving response.

## Abnormal Response

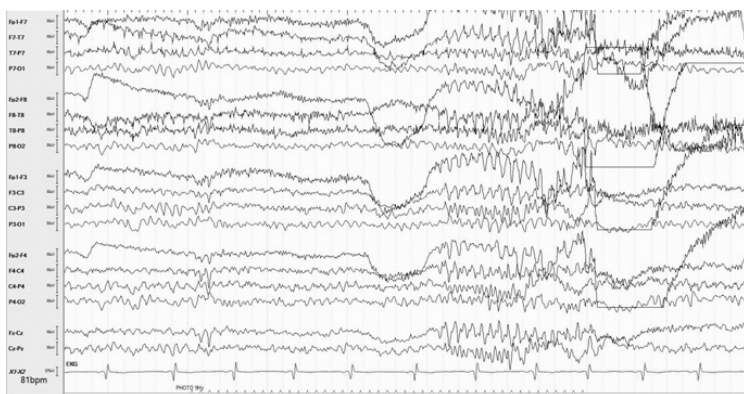
1. *Photo-Paroxysmal Response (PPR)* consists of induction of epileptic discharges in response to photic stimulation. The discharges may continue despite stopping the stimulation (self-sustaining response) or result in a seizure (photo convulsive response). Self-sustaining and self-limited responses have similar associations with epilepsy.

*Significance:* Photo-paroxysmal responses are typically observed between 10 and 30 Hz. They are more common in young women. Eye closure during the stimulation enhances the response while drowsiness, non-REM sleep,

and medications such as Valproate may blunt it. Generalized responses are more common (e.g., JME, CAE, and JAE), but focal responses can also occur (idiopathic occipital epilepsy). Colors and patterns may also enhance the response. Figure 17.6 shows a generalized PPR (non-self-sustaining),

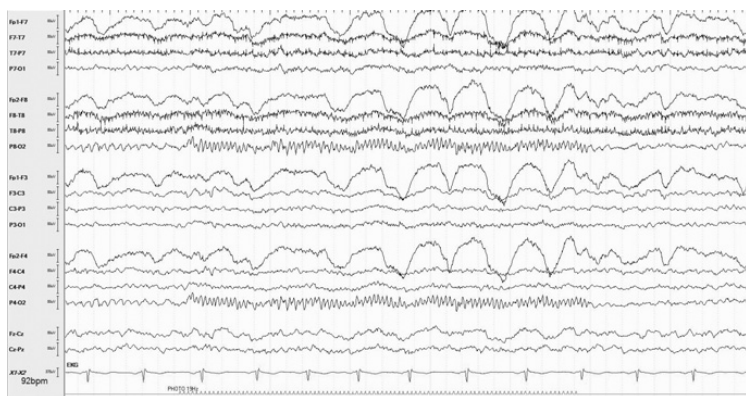


**Figure 17.5** Thirty-two-year-old woman with migraines, normal driving response.



**Figure 17.6** Twenty-seven-year-old woman with JME, photo-paroxysmal response at 9 Hz flash frequency characterized by frontally predominant rhythmic sharp discharges. Provoked discharges do not sustain beyond the photic stimulation.





**Figure 17.9** Twenty-three-year-old woman with migraines, asymmetric driving over the right occipital region.

2. An asymmetric driving response is not always abnormal, especially if present in isolation. It may suggest a structural focus on the absent side. The presence of a driving response in a blind individual may suggest psychogenic or cortical blindness. Figure 17.9 shows an asymmetric photic driving.
3. High-amplitude occipital spikes may be seen with progressive myoclonic epilepsies such as neuronal ceroid lipofuscinosis (NCL); these usually occur at lower flash frequencies (1 or 2 Hz) [3,4].

## Artifactual Responses

These responses are the result of either physiological or external artifact. They have no pathological significance but may be confused with an abnormal response. These are commonly seen in the anterior channels (e.g., frontopolar).

1. Photo-myogenic (myoclonic) response occurs with prominent fast spiky muscle artifact in the representing facial muscle contractions time locked to the photic flash. Figure 17.10 shows photo myoclonic response.
2. Eye flutter artifact may also be seen with photic stimulation. Figure 17.9 shows eye flutter artifact in addition to asymmetric driving response.
3. Electrode artifact may occur in the frontal-polar channels, this may be synchronous with the photic flash (flash artifact).
4. Photic cell artifact (photovoltaic response) is like flash artifact but stops if the electrode is covered. Figure 17.11 shows a photocell artifact.



**Figure 17.10** Photo myoclonic artifact at 8 Hz flash frequency.



**Figure 17.11** Photovoltaic cell artifact (F4) at 17 Hz flash frequency.

## Chapter Summary

1. Activation procedures are used to elicit epileptic activity. Hyperventilation and photic stimulation are commonly used activation procedures.
2. Hyperventilation is contraindicated in cerebrovascular disease.
3. The normal hyperventilation response consists of a high-amplitude, frontally dominant, generalized slowing (called buildup). This is age dependent and may be absent in older patients.
4. Hyperventilation provokes absence seizures in childhood absence epilepsy (CAE). Prolonged buildup may be seen in Moya-Moya disease.
5. Photic stimulation normally results in symmetric, occipital-dominant, repetitive sharps at the flash frequency or a slower harmonic (called driving).
6. Abnormal photic responses include the photo-paroxysmal response which consists of induction of epileptic discharges with photic stimulation and the photo convulsive response results from a seizure elicited during photic stimulation.

## References

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2. Sunder TR, Erwin CW, Dubois PJ. Hyperventilation induced abnormalities in the electroencephalogram of children with moyamoya disease. *Electroencephalography and Clinical Neurophysiology*. 1980 Aug 1;**49**(3–4):414–20.
3. Kasteleijn-Nolst Trenité D, Rubboli G, Hirsch E, et al. Methodology of photic stimulation revisited: updated European algorithm for visual stimulation in the EEG laboratory. *Epilepsia*. 2012 Jan;**53**(1):16–24.
4. Elleder M, Franc J, Kraus J, et al. Neuronal ceroid lipofuscinosis in the Czech Republic: analysis of 57 cases. Report of the “Prague NCL group.” *European Journal of Paediatric Neurology*. 1997 Jan 1;**1**(4):109–14.